

Answer Key & Teacher Notes

Session 1 – Absolute Value as Distance · Session 2 – Absolute Value as Total Distance

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Absolute Value as Distance

TALK PROMPTS

- Why can two different temperatures have exactly the same absolute value?
- Why would a bank care about the absolute value of a negative balance?

WATCH FOR

- Writing a negative absolute value ($|-5.2| = -5.2$).
- Thinking a negative number is closer to 0 because it is “smaller.”
- Mixing up “greater” with “farther from 0” (-36 vs 22).

PARTNER TASK

Two Accounts. One partner plots -45 and 12 on a number line; the other writes both absolute values and explains which balance is closer to zero. Switch roles with -30 and 25 : which number is greater, and which is farther from 0?

SESSION 2 • TEACHER NOTES

Absolute Value as Total Distance

TALK PROMPTS

- Why does zero sitting between the two readings change how you find the distance?
- Why do you join the two distances instead of taking one away from the other?

WATCH FOR

- Subtracting absolute values when 0 is between the readings (-9 to 24 written as 15).
- Adding absolute values when both readings are on the same side of 0 (-18 to -41 written as 59).
- Giving a negative distance.

PARTNER TASK

Peak and Trench. One partner finds the absolute values of -4.7 and 1.8 ; the other traces the trip on a number line and finds the total distance (6.5 km). Switch roles with a Drone Climb from 8 m to 23 m (15 m) and compare the two rules.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.NS.C.7c	1	2	2	1	1	2	2	3	3	1	1
S1 • Form B	6.NS.C.7c	1	2	2	1	1	2	2	3	3	1	1
S2 • Form A	6.NS.C.7c	1	2	1	1	2	2	2	3	3	1	1
S2 • Form B	6.NS.C.7c	1	2	1	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

absolute value: How many units a number is from 0.

distance: How many units apart two points are. It is never **negative**.

magnitude: How big a number is, ignoring its **sign**.

- Absolute value is never **negative**, because distance is never negative.
- Closer to 0 means a **smaller** absolute value.

Example: -13 is **13** units left of 0, so $|-13| = 13$. 6 is 6 units right of 0, so $|6| = 6$. Answer: $13 > 6$, so **-13** is farther from 0.

YOU TRY

1. a. 21 b. 4.5
2. -2.5 , because $|-2.5| = 2.5$ and $2.5 < 3$

EXIT TICKET

1. a. 11 b. 7.25
2. the diver, because $|-12| = 12$ and $12 > 9$
3. **A** (-15)

PRACTICE FORM A

1. a. 16 b. 9 c. 0 d. 2.75
2. a. 6, 2.5, 0.5, 8 b. closest: C (-0.5); farthest: D (8)
3. a. 45 and 12 b. the 12-dollar account
4. 7.5 and 7.5; no, they are on opposite sides of 0
5. 14; the reading is 14 degrees away from 0
6. -36 , because $|-36| = 36$ and $36 > 22$
7. No; $|-5.2| = 5.2$. Absolute value is a distance, and a distance is never negative.
8. a. 85, 40, 120 b. -120 c. 40; greatest means farthest right, not farthest from 0
9. Closeness depends on distance, not on which number is less. $|-19| = 19$ and $|6| = 6$. Since $6 < 19$, the number 6 is closer to 0.
10. **B** (17)
11. **C** (-3)

Self-check word MEASURE

PRACTICE FORM B

1. a. 19 b. 12 c. 0 d. 3.8
2. a. 4, 1.5, 7, 0.5 b. closest: D (0.5); farthest: C (-7)
3. a. 30 and 18 b. the 18-dollar account
4. 6.5 and 6.5; no, they are on opposite sides of 0
5. 9; the reading is 9 degrees away from 0
6. 31, because $|31| = 31$ and $31 > 27$
7. No; $|-8.4| = 8.4$. Absolute value is a distance, and a distance is never negative.
8. a. 95, 60, 150 b. -150 c. 60; greatest means farthest right, not farthest from 0
9. Tessa confused "less than" with "closer to 0." $|-6| = 6$ and $|4| = 4$. Since $4 < 6$, the number 4 is closer to 0.
10. **B** (8.5)
11. **C** (-2)

Self-check word NUMERAL

GUIDED NOTES

absolute value: How many units a number is from 0.

distance: How many units **apart** two points are.

magnitude: The **size** of a number, ignoring its sign.

- If two numbers are on **opposite** sides of 0, add their absolute values.
- If they are on the **same** side of 0, subtract the smaller absolute value from the larger.

Example: 0 is between them, so they are on **opposite** sides. $|-8| = 8$ and $|3| = 3$. Answer: $8 + 3 = 11$ blocks

YOU TRY

1. a. 14 b. 8
2. $30 - 12 = 18$ m

EXIT TICKET

1. a. 17 b. 11
2. $5 + 13 = 18$ degrees
3. **B** (7)

PRACTICE FORM A

1. a. 22 b. 16 c. 15 d. 10
2. a. 12.5 and 7.5 b. $12.5 + 7.5 = 20$ ft
3. $4 + 11 = 15$ floors
4. $19 - 6 = 13$ m
5. $2.4 + 3.6 = 6$ km
6. $41 - 18 = 23$ m; both readings are on the same side of 0, so subtract
7. $67 + 33 = 100$ dollars
8. a. $120 + 340 = 460$ m b. $340 - 90 = 250$ m c. Trail 1 starts below 0, so it climbs 120 m just to reach 0.
9. 0 is between -9 and 24, so the readings are on opposite sides. Hugo skipped the 9 degrees from -9 up to 0. Add: $9 + 24 = 33$ degrees.
10. **B** (15)
11. **B** (add their absolute values)

Self-check word DISTANCE

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • ABSOLUTE VALUE AS DISTANCE

1. a. 14 b. 8 c. 0 d. 2.5 e. $\frac{3}{4}$ f. $1\frac{1}{2}$
2. a. $|A| = |-7| = 7$ b. $|B| = |5| = 5$ c. point A
3. a. $>$ ($9 > 7$) b. $=$ ($4 = 4$) c. $<$ ($-10 < 3$) d. $=$ ($6.5 = 6.5$) e. $<$ ($12 < 15$) f. $<$ ($0 < 1$)
4. a. $|-35| = 35$ and $|20| = 20$ b. Rosa's ($35 > 20$)
5. $-2, -1, 0, 1, 2$ (each is less than 3 units from 0)
6. 4.5 and -4.5 . Both are 4.5 units from 0, on different sides.

PRACTICE FORM B

1. a. 23 b. 18 c. 14 d. 11
2. a. 10.5 and 7.5 b. $10.5 + 7.5 = 18$ degrees
3. $6 + 8 = 14$ floors
4. $23 - 8 = 15$ m
5. $1.8 + 4.7 = 6.5$ km
6. $47 - 12 = 35$ m; both readings are on the same side of 0, so subtract
7. $48 + 27 = 75$ dollars
8. a. $150 + 280 = 430$ m b. $280 - 60 = 220$ m c. Trail 1 starts below 0, so it climbs 150 m just to reach 0.
9. 0 is between -7 and 10, so the floors are on opposite sides. Malik should add the distances: $7 + 10 = 17$ floors.
10. **B** (15)
11. **C** (10)

Self-check word ABSOLUTE

SHEET 2 • ABSOLUTE VALUE AS TOTAL DISTANCE

1. a. $5 + 6 = 11$ b. $1 + 6 = 7$ c. $5 - 1 = 4$
2. a. 10 b. 8 c. 6.5
3. $8 + 15 = 23$ degrees
4. -3 and 9 are on opposite sides of 0: $3 + 9 = 12$ floors
5. -9 and 5 are on opposite sides of 0, so he should add: $9 + 5 = 14$ units.
6. **B** (19)
7. **B** (38 feet)